

# **DEEP LEARNING REPORT**

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#### **Section A: Ablation Analysis**

##### **A.1 Patch Size**

I experimented with three different patch sizes: 2, 4, and 8. From the results in the ablation logs, I observed that the model with patch size 4 achieved a validation accuracy of 0.4929, which is the baseline. The model with patch size 2 performed slightly better than patch size 4, while the model with patch size 8 had the lowest accuracy.

The main difference between patch sizes is the number of tokens  $N$ . When the patch size is smaller, the number of patches increases. For example, with patch size 2, the image is divided into many small patches, so the model gets more information which increases the model's ability to capture fine features that improves accuracy.

Anyways, smaller patches also increase computation and make training harder. With too many tokens, the model becomes more complex and may not learn efficiently. This is why patch size 2 does not show a big improvement over patch size 4. Whereas, with patch size 8, the number of patches is very small. This decreases the amount of spatial detail available to the model, which leads to a very lower accuracy. Finally, I noticed that patch size 4 gives a good balance between model performance and computational cost.

##### **A.2 Number of Layers**

I compared models with 2, 4, and 8 Transformer layers. The model with 4 layers achieved the best performance with a validation accuracy of 0.4929. The model with 2 layers had lower accuracy and increasing the depth to 8 layers had not improve performance.

This shows that accuracy does not increase steadily with more layers. On this small dataset, adding more layers beyond a certain point does not help and may even make training harder.

There are a few reasons for this that is first, deeper models are more difficult to train and may not converge well. Second, the dataset is relatively small, so a very deep model may overfit or fail to generalize. Third, the model size increases with depth, but the benefit from extra layers becomes smaller. From this experiment, we can say that a moderate depth, like 4 layers which works the best

### **A.3 Positional Encoding**

I compared the baseline model with the model trained without positional encoding using the results from no\_pos\_enc.json. The baseline model achieved a final validation accuracy of 0.4929, while the model without positional encoding achieved a lower accuracy of 0.4602.

Looking at the training curve, both models show a similar trend where accuracy increases over time. However, the model without positional encoding consistently performs worse across all epochs and ends with a lower final accuracy. This shows that removing positional information negatively affects learning.

This happens because Vision Transformers do not have any built-in understanding of spatial relationships. Without positional encoding, the model treats all image patches as an unordered set. In other words, it cannot tell where a patch comes from in the image. Since image classification depends heavily on spatial structure (for example, where objects are located), the model struggles when positional information is removed. As a result, its performance drops compared to the baseline.

## **Section B: Quantitative Analysis Interpretation**

### **B.1 Attention Entropy**

The attention entropy values from the analysis are:

- Layer 0: 4.1993
- Layer 1: 4.0315
- Layer 2: 4.0833

- Layer 3: 4.6592

The entropy does not show a clear increasing or decreasing trend with depth. It decreases a little from layer 0 to layer 1 then it remains similar in layer 2, and then increases in layer 3. This means that attention does not become more focused as we go deeper into the network. Instead, the last layer has higher entropy, which indicates that attention is more spread out across many patches. This shows that earlier layers may focus on more specific regions of the image, while deeper layers combine information from multiple regions.

## **B.2 Positional Embedding Correlation**

The Pearson correlation between positional similarity and spatial distance is:

$$r = -0.2592$$

The correlation is negative, which means that patches that are closer together in the image tend to have more similar positional embeddings. This is expected because nearby patches usually represent related parts of the image. The magnitude of the correlation is moderate, which shows that the model has learned some spatial structure, but not perfectly. This is reasonable because positional embeddings are learned in a 1D sequence, not directly from 2D coordinates. This result shows that the model has learned spatially coherent positional representations.

## **B.3 Per-Class Accuracy and Confusions**

The per-class accuracies are:

- Class 0: 0.54
- Class 1: 0.589
- Class 2: 0.348
- Class 3: 0.298
- Class 4: 0.413
- Class 5: 0.432
- Class 6: 0.607
- Class 7: 0.528
- Class 8: 0.64
- Class 9: 0.534

The model performs best on class 8 (0.64) and worst on class 3 (0.298). The top 3 confusion pairs are:

cat -> dog (244), truck -> automobile (216), automobile -> truck (207)

The hardest classes for the model are those that are visually similar. For example, cats and dogs have similar shapes and textures, especially in low-resolution images like CIFAR-10. Similarly, trucks and automobiles share common features such as wheels and body structure. Because of these similarities, the model finds it difficult to distinguish between these classes, leading to higher confusion.

#### **B.4 Mean Attention Distance**

The mean attention distance values are:

- Layer 0: 4.0992
- Layer 1: 4.0919
- Layer 2: 4.2073
- Layer 3: 4.2156

The attention distance increases slightly as we move to deeper layers. This means that the model attends to patches that are farther apart in deeper layers. In other words, the effective receptive field grows with depth. This behavior is similar to convolutional neural networks, where deeper layers capture more global features. But, in Vision Transformers, this happens through attention instead of fixed convolutional filters. This shows that the model starts by focusing on local regions and gradually shifts to capturing more global information.